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PAPERS (SHORT
AUTHORLIST)

* denotes joint first-author papers, † denotes student supervised.

34. †Avinash Tiwari, **Aditya Vijaykumar**, Shasvath J. Kapadia, Sourav Chatterjee
Identifying and characterizing extragalactic circum-CBC exoplanets with future gravitational-wave detectors
[arXiv:2607.09658](#).
In press: Astrobites
33. Avinash Tiwari, Shasvath J. Kapadia, **Aditya Vijaykumar**, Sourav Chatterjee
Periodic line-of-sight velocity-driven modulations to gravitational waves emitted by compact binaries in Keplerian outer orbits
[arXiv:2607.09644](#).
32. **Aditya Vijaykumar**, Reed Essick
Semianalytic Sensitivity Estimates for Out-of-Bank Gravitational-Wave Signals
[arXiv:2606.14045](#).
31. Maya Fishbach et al. (including **Aditya Vijaykumar**)
Implications of Low Neutron Star Merger Rates for Gamma-Ray Bursts, r-process Production, and Galactic Double Neutron Stars
[ApJL 1006, L49](#), [arXiv:2604.05059](#).
30. †Neha Sharma, **Aditya Vijaykumar**, Prayush Kumar
Rapid inference of gravitational-wave signals in the time domain using a heterodyned likelihood
[Phys. Rev. D 114, 044007](#), [arXiv:2601.11239](#).
29. **Aditya Vijaykumar**, Amanda M. Farah, Maya Fishbach
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28. Amanda M. Farah, **Aditya Vijaykumar**, Maya Fishbach
The steep redshift evolution of the hierarchical binary black hole merger rate may cause the z - χ_{eff} correlation
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27. N.V. Krishnendu, Tamara Evstafyeva, ***Aditya Vijaykumar**, William E. East et al.
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26. Madison VanWyngarden, Maya Fishbach, **Aditya Vijaykumar**, Alexandra G. Guerrero, Daniel E. Holz
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25. Hui Tong et al. (including **Aditya Vijaykumar**)
Evidence of the pair instability gap in the distribution of black hole masses
[Nature 652, 874](#), [arXiv:2509.04151](#).
24. Colm Talbot et al. (including **Aditya Vijaykumar**)
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23. Avinash Tiwari, Prolay Chanda, Shasvath J. Kapadia, Susmita Adhikari, **Aditya Vijaykumar**, Basudeb Dasgupta
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22. Andris Doroszmai, Isobel M. Romero-Shaw, **Aditya Vijaykumar**, Silvia Toonen, et al.
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21. [†]Avinash Tiwari, **Aditya Vijaykumar**, Shasvath J. Kapadia, Shrobana Ghosh, Alex B. Nielsen
A pipeline to search for signatures of line-of-sight acceleration in gravitational wave signals produced by compact binary coalescences
[Phys. Rev. D 113, 104072](#), [arXiv:2506.22272](#).
20. Kanchan Soni, **Aditya Vijaykumar**, Sanjit Mitra
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19. [†]Avinash Tiwari, **Aditya Vijaykumar**, Shasvath J. Kapadia, Sourav Chatterjee, Giacomo Fragione
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18. Alexandra G. Hanselman, **Aditya Vijaykumar**, Maya Fishbach, Daniel E. Holz
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15. Mukesh Kumar Singh, Shasvath J. Kapadia **Aditya Vijaykumar**, Parameswaran Ajith
Impact of higher harmonics of gravitational radiation on the population inference of binary black holes
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14. [†]Kruthi Krishna, ***Aditya Vijaykumar**, Apratim Ganguly, et al
Accelerated parameter estimation in Bilby with relative binning
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13. **Aditya Vijaykumar**, Maya Fishbach, Susmita Adhikari, Daniel E. Holz
Inferring host galaxy properties of LIGO-Virgo-KAGRA's black holes
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12. Divyajyoti, N.V. Krishnendu, Muhammed Saleem, Marta Colleoni, **Aditya Vijaykumar**, K.G. Arun, Chandra Kant Mishra
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11. [†]Avinash Tiwari, ***Aditya Vijaykumar**, Shasvath J. Kapadia, Giacomo Fragione, Sourav Chatterjee
Accelerated binary black holes in globular clusters: forecasts and detectability in the era of space-based gravitational-wave detectors
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10. **Aditya Vijaykumar**, [†]Avinash Tiwari, Shasvath J. Kapadia, K.G. Arun, Parameswaran Ajith
Waltzing binaries: Probing line-of-sight acceleration of merging compact objects with gravitational waves
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9. [†]Adhrit Ravichandran, **Aditya Vijaykumar**, Shasvath J. Kapadia, Prayush Kumar
Rapid Identification and Classification of Eccentric Gravitational Wave Inspirals with Machine Learning
Submitted to PRD, [arXiv:2302.00666](#).

8. Srashti Goyal, **Aditya Vijaykumar**, Jose Maria Ezquiaga, Miguel Zumalacarregui
Probing lens-induced gravitational-wave birefringence as a test of general relativity
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7. Bikram Keshari Pradhan, **Aditya Vijaykumar**, Debarati Chatterjee
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6. **Aditya Vijaykumar**, Shasvath J. Kapadia, Parameswaran Ajith
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4. Sumit Kumar, **Aditya Vijaykumar**, Alexander H. Nitz
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3. M. Saleem, Javed Rana, V. Gayathri, ***Aditya Vijaykumar** et al.
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2. **Aditya Vijaykumar**, M. V. S. Saketh, Sumit Kumar, Parameswaran Ajith, Tirthankar Roy Choudhury
Probing the large scale structure using gravitational wave observations of binary black holes,
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1. **Aditya Vijaykumar**, Shasvath J. Kapadia, Parameswaran Ajith
Constraints on the time variation of the gravitational constant using gravitational wave observations of binary neutron stars,
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PAPERS (LONG
AUTHORLIST,
WITH
SUBSTANTIAL
CONTRIBUTION)

11. Abac et al. (LIGO Scientific, Virgo, and KAGRA Collaborations)
GWTC-5.0: Population Properties of Merging Compact Binaries,
[arXiv:2605.27226](#).
10. Abac et al. (LIGO Scientific, Virgo, and KAGRA Collaborations)
GWTC-5.0: Observations from the Second Part of the Fourth LIGO-Virgo-KAGRA Observing Run and Updates to the Gravitational-Wave Transient Catalog,
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9. Abac et al. (LIGO Scientific, Virgo, and KAGRA Collaborations)
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8. Abac et al. (LIGO Scientific, Virgo, and KAGRA Collaborations)
Upper Limits on the Isotropic Gravitational-Wave Background from the first part of LIGO, Virgo, and KAGRA's fourth Observing Run,
Phys. Rev. D **114**, 042001, [arXiv:2508.20721](#).
7. Abac et al. (LIGO Scientific, Virgo, and KAGRA Collaborations) [**Paper Writing Team Lead**]
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6. Abac et al. (LIGO Scientific, Virgo, and KAGRA Collaborations)
GWTC-4.0: Updating the Gravitational-Wave Transient Catalog with Observations from the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run,
[ApJL 1004, L22](#), [arXiv:2508.18082](#).
5. Abac et al. (LIGO Scientific, Virgo, and KAGRA Collaborations)
GW231123: a Binary Black Hole Merger with Total Mass 190-265 $M_{\odot}$,
[ApJL 993, L25](#), [arXiv:2507.08219](#).
4. Abbott et al. (LIGO Scientific and Virgo Collaborations)
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[Phys. Rev. X 11, 021053](#), [arXiv:2010.14527](#).
1. P. Virtanen et al. (including **Aditya Vijaykumar** as *SciPy 1.0 Contributor*)
SciPy 1.0—Fundamental Algorithms for Scientific Computing in Python,
[Nat. Methods 17, 261](#), [arXiv:1907.10121](#).